18

Lucuma (*Pouteria lucuma* (Ruiz and Pav.) Kuntze)

E. M. Yahia and F. Guttierrez-Orozco, Autonomous University of Queretaro, Mexico

Abstract: Lucuma fruit is a good source of fiber, minerals, β-carotene, phenolics and niacin. The fruit is consumed mostly in Peru and Chile, although it is also known in a few other countries, and mostly in processed form in ice creams, bakery products and preserves. Almost no postharvest information is available on this fruit, and research is needed on nearly all aspects of fruit physiology and handling. This brief chapter is intended to report the information available on this fruit and to draw attention to the research that needs to be carried out and the information that needs to be generated.

Key words: *Pouteria lúcuma*, postharvest, nutrition, processing, handling.

18.1 Introduction

The lucuma tree is not a tropical plant, rather it grows at temperate elevations in dry locations. Native to Peru, it is a favorite ice cream flavor. Most of the crop is usually used in dehydrated or frozen form since the soft flesh of the fresh fruit is easily damaged, making transportation difficult.

18.1.1 Origin, botany, morphology and structure

Lucuma (*Pouteria lúcuma*) belongs to the Sapotaceae family and is also known as lucma, lucmo, lúcuma, lúcumo, mammon, cumala, rucma or marco. It is a native fruit of the highlands of Peru, Ecuador, and Chile and was an important part of the pre-hispanic diet of the people from these areas. It grows well in Mexico and Hawaii but the fruit is not very popular in these areas.

The lucuma tree is an evergreen with a height of 8 to 15 m and a dense crown with branches that produce white latex. It has a long juvenile period of about

15 years. It is propagated by grafting scions onto seedling rootstocks, but this method causes high variability in production. Propagating leafy cuttings under mist or in a plastic hermetic chamber can also be successful (Duarte, 1990). *In vitro* propagation has been attempted by using shoot tips, but the majority of the plants died after transfer to greenhouse conditions (Jordan and Oyanedel, 1992). On the other hand, micropropagation of lucuma plants combined with inoculation with arbuscular mycorrhizal fungi has been shown to improve their growth and development (Padilla *et al.*, 2006).

Lucuma fruit has an ovoid to elliptical shape with a pointed or depressed apex. (See Plate XXXII in the color section between pages 274 and 275.) It is 7.5–10 cm in size with thin skin which is greenish-yellow when the fruit is fully ripe. The flesh is dry, with a starchy orange-yellow color and pumpkin-like sweet flavor. The unripe fruit contains latex. Often, two seeds are found, although 1–5 are possible. They are round to oval in shape, dark brown with a glossy appearance and a white hilum. Germination of lucuma seeds is affected by dessication (below 19% matter content), thus a less drastic dessication process should be used when storing lucuma seeds (Magne Ojeda *et al.*, 2005). The most common varieties are 'Seda' and 'Palo'. The former is mainly consumed fresh due to the higher water content, while the latter is used to make ice cream.

18.1.2 Worldwide importance

Peru is the main producer of lucuma (88% of world production), although production in Chile has been increasing (12%). Introduction of this species to the US has not been successful, especially because the lucuma tree is very sensitive to freezing temperatures. In 2010, lucuma from Peru was mainly exported to Chile (74%), followed by the US and Canada (AMPEX, 2010). It is almost unknown outside these areas, although it can be found in countries like Costa Rica, Mexico and Hawaii. The fruit is exported mainly as frozen pulp (79%) and flour, which are used in bakery products, ice cream and jams (AMPEX, 2010).

18.1.3 Culinary uses, nutritional value and health benefits

The lucuma can be eaten raw, although some people find the raw fruit not very appealing since it has an odd aftertaste. The pulp is made into preserves or used in ice cream, yogurt, many desserts or bakery products, for example in pastries and as a cookie filling. Lucuma flavor ice cream is very popular in Peru. Since lucuma gives a sweet taste to the foods to which it is added, it is a healthy natural alternative to sweeteners. In spite of the sweet taste, lucuma has a low sugar concentration. Lucuma fruits are a good source of fiber, vitamins and minerals. Fiber in lucuma is mainly found in the insoluble form (Glorio *et al.*, 2008). High concentrations of β-carotene, niacin and iron have been found in the fruit. Some of the sugars present in lucuma fruit are glucose, fructose, sucrose and inositol in the following amounts: 8.4, 4.7, 1.7, and 0.06 g, respectively (Herbal Guides, 2010). The nutritional value of lucuma fruit is presented in Table 18.1.

Table 18.1 Nutritional value of lucuma fruit (per 100 g of fruit)

Constituent	Approximate value
Water content	62 %
Calories	143.8
Protein	2.3 g
Carbohydrates	33.2 g
Fat	0.2 g
Fiber	1.1 g
Calcium	16 mg
Phosphorus	26 mg
Iron	0.4 mg
Thiamin	0.01 mg
Riboflavin	0.14 mg
Niacin	1.96 mg
Vitamin C	5.4 mg

Source: http://www.guallarauco.cl (2010)

The antioxidant capacity of lucuma extracts was found to be high. Catechin and epicatechin, present in these extracts, may contribute to the observed antioxidant capacity (Ma, 2004). A recent study found aqueous extracts of lucuma to have the highest concentration of phenolic compounds (11.4 mg g^{-1} dw) when compared to other Peruvian fruits and a high α-glucosidase inhibitory activity. The latter could suggest lucuma as a food-based treatment to complement diabetes management (Silva Pinto *et al.*, 2009). The growth of *Staphylococcus aureus* was inhibited by extracts of lucuma (Lazo, 1990). Lupeol and a and β-amyrin in the form of fatty acid esters and acetates, as well as the cyanogenic glycoside lucumin, have been identified in seeds of lucuma from Belize (Merfort, 1984).

18.2 Fruit development and postharvest physiology

18.2.1 Fruit growth, development and maturation

The growth of lucuma fruits is sigmoidal and is accelerated by higher temperatures (Sandoval, 1997).

18.2.2 Respiration, ethylene production and ripening

Lucuma is a climacteric fruit according to its CO_2 production pattern (Yahia, 2004). Ripening of the fruit includes changes in color from green to yellow, loss of firmness and an increase in soluble solids. Intense respiration and sugar accumulation are characteristic during ripening of lucuma (Lizana *et al.*, 1986; Yahia, 2004).

18.3 Maturity and quality components and indices

One of the most common maturity indices used for lucuma is the change in skin color from green to yellow, although pulp color can vary from green to yellowish green and light yellow to orange-yellow color (Lizana, 1980). Alternatively, soluble solids content may be used as a maturity index. However, because of the low water content and density of the pulp, it is necessary to homogenize it in water in order to disrupt the pulp and get an accurate value (Lizana, *et al.*, 1986).

Based on peel and pulp color, texture, soluble solids content and respiration, five maturity stages have been developed (Table 18.2). Fruit of lucuma var. 'Palo' have 0.11% acidity and 8°Brix at maturity (Glorio *et al.*, 2008).

Table 18.2 Classification of lucuma fruit into different maturity stages on the basis of peel and pulp color

Class	Peel color	Pulp color
1	Light yellow	Light yellow
2	Light green	Creamy yellow
3	Yellow-green	Yellow
4	Green-yellow	Dark yellow
5	Green-yellow	Orange-yellow

Source: Lizana (1980)

18.4 Postharvest handling factors affecting quality

18.4.1 Temperature management

The quality of lucuma fruit stored at 7 °C is not affected when the storage duration is up to 7 days. After longer periods of storage, fruit do not ripen uniformly. If stored at 13 and 18 °C, fruit can be kept for up to 14 days before showing signs of decay (Sandoval, 1997).

18.4.2 Physical damage

The soft texture of lucuma fruit makes it very prone to physical damage (Plate XXXIIC), and because of that, lucuma is commercialized as frozen pulp or flour.

18.4.3 Water loss

Lucuma fruit is highly sensitive to water loss postharvest (Sandoval, 1997).

18.5 Physiological disorders

As mentioned above, storage of lucuma fruit at 7 °C for more than 7 days negatively affects fruit ripening and quality (Sandoval, 1997).

18.6 Insect pests and their control

As the lucuma tree is only affected by a few pests, it is a good candidate for organic production. Trees are sometimes periodically washed with pure water to keep them free of pests.

18.7 Postharvest handling practices

18.7.1 Harvest operations

Lucuma trees start producing fruit after 4 or 5 years and provide fruit year-round. It is common to see ten-year-old trees producing 200-300 fruit per year (Prolucuma, 2010). Although mature fruit fall from the tree they still need to ripen for several days before they can be consumed.

18.7.2 Control of ripening and senescence

Irradiation of lucuma (5–100 × 10^3 rad) barely affects the respiratory rate of the fruit and the shelf life is not significantly extended. Chemical parameters such as total sugars, water content, ash and vitamin C are not affected by irradiation, except that acidity increases slightly. Irradiation treatment at higher than 5000 rad causes loss of quality, making the fruit unacceptable for consumption. A strong fungicidal effect is seen when fruit are treated with more than 5000 rad (Díaz *et al.*, 1969).

18.7.3 Recommended storage and shipping conditions

Storage at low temperatures for more than 7 days negatively affects ripening. Temperatures of 13 or 18 °C can be used to store lucuma fruit for up to 14 days. Due to its high sensitivity to water loss, the fruit need to be kept at high relative humidity (Sandoval, 1997). Modified atmosphere (especially for packaging) could be helpful in maintaining the quality of fresh fruit (Yahia, 1998; 2008).

18.8 Processing

Lucuma products available in the market include puree and pulp (Plate XXXIID). These products are made from fruit that have been washed, disinfected, peeled and seeded. The pulp is ground, vacuum-packed and quick frozen at −25 °C. This way the pulp is stable for 2 years without significant changes in quality. This product is used in drinks, ice cream and baking. Lucuma jam is made by mixing the pulp with cooked sugar; the product is vacuum-packed in polyethylene (PE) bags inside corrugated cardboard boxes of 20 kg each. Pulp processed in this way is stable for 1–2 years at −18 °C (Guallarauco, 2010; Prolucuma, 2010).

Freeze-dried pulp is also available. Freeze-drying preserves the flavor characteristics better than dehydration. The fruit is washed, disinfected, peeled, seeded and cut before being frozen, lyophilized and ground. The final product is packed in PE bags of 40 kg (Prolucuma, 2010).

Flour produced from dehydrated lucuma fruit is used as a flavoring agent in ice creams or dairy products. The fruit is selected, disinfected, peeled, seeded and cut, before being dried at 60 °C in hot air tunnels. Lucuma flour is packed in 10-kg bags (Prolucuma, 2010).

18.9 Conclusions

Although fresh lucuma is little known outside its area of origin, processed products such as flour and frozen pulp are available in different markets. Handling of fresh lucuma is difficult because the fruit is very prone to physical damage. The high content of some nutrients such as β-carotene, niacin and iron makes lucuma a good choice especially in areas where deficiency of these nutrients is frequent.

18.10 References

Asociación Macroregional de Productores para la Exportación (AMPEX) (2010), Boletin Informativo Estadistico 2009-VI. Available from: http://www.ampex.com.pe [Accessed 28 June 2010].

Díaz D, Behrens J E and Recke W W (1969), Efecto de las radiaciones gama sobre la maduración y conservación de lúcumas y chirimoyas, *Revista de la Real Academia de Ciencias*, 311–321.

Duarte O (1990), Propagación del lúcumo (*Lucuma obovata* H.B.K.) por estacas terminales con hojas, *Acta Horticulturae*, **310**, 235–238.

Glorio P, Repo-Carrasco R, Velezmoro C *et al.* (2008), Fibra dietaria en variedades peruanas de frutas, tuberculos, cereales y leguminosas, *Rev Soc Quim Peru*, **74** (1), 46–56.

Guallarauco Web Site. Agro-industrial product processing, packing and marketing company. Colina, Santiago, Chile. Available from: http://www.guallarauco.cl [Accessed 26 February 2010].

Herbal Guides (2010), Available from http://herbalguides.com/guides/lucuma [Accessed 20 February 2010].

Jordan M and Oyanedel E (1992), Regeneration of *Pouteria lucuma* (*Sapotaceae*) plants in vitro, *Plant Cell Tiss Organ Cult*, **31**, 249–252.

Lazo W (1990), Antimicrobial action of some plants of medicinal use in Chile, *Boletín Micológico*, **5**(1–2), 25–28.

Lizana, L A (1980), Lucuma. pp. 373–380. In: *Fruits of Tropical and Subtropical Origin*. Nagy S, Shaw P E and Wardowski W F (eds). Florida Science Source, Inc., Lake Alfred, FL.

Lizana L A, Reginato G and Lavanderos J C (1986), Maturation characteristics in lucuma (*Lucuma bifera* Mol., *L. obovata* HBK), *Proceedings of the Interamerican Society for Tropical Horticulture*, **30**, 105–120.

Ma J (2004), Polyphenolic antioxidants from Sapotaceae fruits. Thesis, The City University of New York. New York, NY. 98 p.

Magne Ojeda J, Gonzalez Saucedo L, Sacande M, Joker D, Dulloo M E and Thomsen K A (2005), Desiccation and storage of lucuma, *Pouteria macrophylla* seeds, International Plant Genetic Resources Institute (IPGRI), Rome, Italy, *Comparative storage biology of tropical tree seeds*, 258–262.

Merfort I (1984), Phytochemical study of *Lucuma mammosa, Fitoterapia*, **55** (5), 316–317.

Morton J (1987), Lucmo. pp. 405–406. In: *Fruits of warm climates*. Julia F. Morton, Miami, FL.

Padilla I M G, Carmona E, Westendorp N and Encina C L (2006), Micropropagation and effects of mycorrhiza and soil bacteria on acclimatization and development of lucumo (*Pouteria lucuma* R. and Pav.) var. La Molina, *In Vitro Cell Dev Biol-Plant*, **42**, 193–196.

Prolucuma Web Site. Asociación de Productores de Lucuma del Perú. Available from: http://www.prolucuma.com [Accessed 16 June 2010].

Sandoval BLH (1997), Caracterización del crecimiento y comportamiento postcosecha de frutos de lucumo (*Pouteria lucuma* (R. et P. O. Kze)), Chile University, Thesis, Record number CL1999000075, AGRIS 2010 – FAO of the United Nations.

Silva Pinto M, Ranilla LG, Apostolidis E, Lajolo FM, Genovese MI and Shetty K (2009), Evaluation of antihyperglycemia and antihypertension potential of native Peruvian fruits using in vitro models, *J Med Food*, **12**(2), 278–291.

Yahia EM, editor (2008), ‘Modified and controlled atmospheres for the storage, transportation, and packaging of horticultural commodities’, Taylor ’ Francis, Boca Raton, FL.

Yahia E M (2004), ‘Sapodilla and related fruits’, in: U.S. Dept. Agric. Agric. Handbook #66 Available from http://www.ba.ars.usda.gov/hb66/index.html [Accessed 6 June 2010].

Yahia EM (1998), Modified and controlled atmospheres for tropical fruits, *Horticultural Reviews*, **22**, 123–183.

Plate XXXI (Chapter 17) Different processed products of loquat fruit: (a) loquat juice; (b) loquat wine; (c) loquat jam; (d) loquat ointment; and (e) loquat tea.

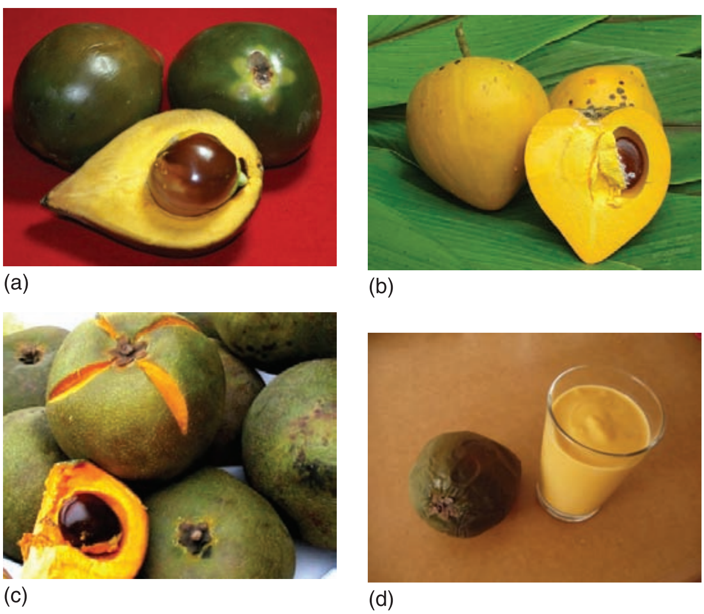

Plate XXXII (Chapter 18) (a) Exterior of the lucuma fruit; (b) interior of the lucuma fruit; (c) Lucuma fruit showing physical injury; (d) Lucuma fruit and juice